ASCII MASTER FLOW — PANEL 1/12: Atom and periodic identity map

```
NUCLEUS -> protons + neutrons
ATOMIC NUMBER -> protons -> element identity
MASS NUMBER -> protons + neutrons
ISOTOPE / ISOBAR / ISOTONE -> different comparison axes
PERIODIC TABLE -> atomic number + recurring properties
```

ASCII MASTER FLOW — PANEL 2/12: Bond and force ladder

```
IONIC -> electron transfer + ion attraction
COVALENT -> shared electron pair
METALLIC -> delocalised electrons
HYDROGEN BOND / VAN DER WAALS -> weaker interparticle forces
TRAP -> do not collapse bond and force categories
```

ASCII MASTER FLOW — PANEL 3/12: Stoichiometric accounting rail

```
WRITE SPECIES -> BALANCE EQUATION
CONVERT SUPPLIED QUANTITIES -> MOLES
COMPARE STOICHIOMETRIC REQUIREMENT
IDENTIFY LIMITING REAGENT -> THEORETICAL PRODUCT
ACTUAL / THEORETICAL -> YIELD, only with supplied data
```

ASCII MASTER FLOW — PANEL 4/12: State and solution matrix

SOLID / LIQUID / GAS -> arrangement + mobility + separation
PHASE CHANGE -> physical identity retained
SOLUTION -> molecular/ionic dispersion
CONCENTRATION -> amount on a stated basis
SOLUBILITY -> equilibrium limit under stated conditions

ASCII MASTER FLOW — PANEL 5/12: Acid base buffer control loop

ACID DONATES H+ <-> BASE ACCEPTS H+

pH -> logarithmic activity measure

NEUTRALISATION -> acidic/basic character reduced

BUFFER -> consumes limited added acid/base

SALT SOLUTION -> acidic / basic / neutral depends on ions

ASCII MASTER FLOW — PANEL 6/12: Electrochemical two-cell map

OXIDATION -> anode | REDUCTION -> cathode
GALVANIC CELL -> chemical energy to electrical energy
ELECTROLYTIC CELL -> electrical energy drives chemistry
CORROSION / PLATING / EXTRACTION -> applied routes
SIGN TRAP -> cell type changes signs, not reaction locations

ASCII MASTER FLOW — PANEL 7/12: Kinetics versus equilibrium

RATE AXIS -> collisions + activation energy
CATALYST -> alternative lower-energy pathway
EQUILIBRIUM AXIS -> forward rate equals reverse rate
CATALYST -> faster approach in both directions
NO SHIFT -> equilibrium position unchanged

ASCII MASTER FLOW — PANEL 8/12: Organic and polymer family tree

FUNCTIONAL GROUPS -> alcohol | aldehyde | ketone | acid | amine | ester
ADDITION POLYMER -> no small-molecule elimination
CONDENSATION POLYMER -> small molecule eliminated
THERMOPLASTIC <-> THERMOSET -> reheating boundary
HYDROGEL -> cross-linked hydrophilic network

ASCII MASTER FLOW — PANEL 9/12: Materials structure-property wheel

```
DIAMOND -> tetrahedral network
GRAPHITE -> layered structure
GRAPHENE / FULLERENE / NANOTUBE -> distinct carbon architectures
ALLOY -> composition + processing alter properties
RULE -> same element can yield different material behaviour
```

ASCII MASTER FLOW — PANEL 10/12: Fuel conversion fault tree

```
COMBUSTION -> oxidation + useful heat + possible pollutants
INCOMPLETE COMBUSTION -> different product burden
GASIFICATION -> controlled oxygen/steam conversion
SYNGAS -> carbon monoxide + hydrogen dominant route
OUTCOME FIREWALL -> product is not efficiency or climate proof
MUST REMEMBER: Chemistry fundamentals link atomic structure, mole/stoichiometry, bonding,...
```
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ASCII MASTER FLOW — PANEL 11/12: Environmental compartment map

```
STRATOSPHERE -> ozone protective role
TROPOSPHERE -> ozone pollutant role
WATER / SOIL -> nutrients, acids, salts, persistent chemicals
EXPOSURE -> concentration + route + duration + conditions
LABEL -> not safety, hazard or regulation evidence
CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent,...
```

```
CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent,...
```

ASCII MASTER FLOW — PANEL 12/12: PYQ and policy evidence ladder

```
WATER POLARITY -> many ionic/polar solutes, not every substance
BPA / TRICLOSAN -> presence differs from exposure and regulation
HYDROGEL -> material class differs from universal performance
PRINCIPLE -> PROCESS -> APPLICATION -> INSTITUTION
PAGE / DOCUMENT -> not deployment, outcome, hazard or law
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...
```
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...